Institut Polytechnique de Paris
Paris Region, France
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Dear Editor,

We are pleased to submit our paper titled "Real-Time Redesign of Public Transport Lines under Fluctuating Traffic Conditions".

Public Transport (PT) design is traditionally based on nominal conditions (e.g., peak or average link travel times). Although conditions stochastically fluctuate, the designed network structure remains mostly unvaried and only limited corrective actions (e.g., stop-skipping, holding) are implemented. As a result, public transport networks struggle to maintain efficiency.

To overcome this limitation, the **key contribution** of our article is a predict-then-optimize framework that proactively redesigns PT lines in real time, accounting for uncertainty in traffic conditions. We realistically model the inevitable prediction errors by fitting error probability distributions on the inferences of well-established traffic prediction models on real traffic datasets. Then, we proactively re-optimize PT lines via multiobjective optimization, based on predicted traffic conditions. Extensive numerical evaluations in Mandl's network and a real-world network in Beijing show that the proposed proactive redesign outperforms traditional static network designs, in terms of user- and operator-oriented metrics.

We believe that our research methodology and our findings are strongly consistent with the aim of Transportation Science as it develops innovative approaches for managing public transportation systems under uncertainty with improved resilience and operational efficiency.

The preliminary version of this work was presented at the 19th Learning and Intelligent Optimization Conference (LION19). However, this article is deeply different than that preliminary version:

- We now realistically model the error distribution resulting from well-established deep learning based prediction models. This allows us to evaluate the effectiveness of our optimization, even when faced with the inevitable inaccuracies of prediction models. In LION19, instead, we made the unrealistic assumption that link travel times are perfectly known in advance.
- To ensure that the proposed real-time network modifications are not excessive, we add in the objective function a Jaccard index, which measures the similarity with respect to the original network. In LION19, a simple count of overlapping links was used. The new formulation, although more accurate, leads to a non-linear optimization problem (while in LION19 it was linear), which requires a more involved solution method.
- We now add a large-scale scenario, namely the Beijing dataset.
- We changed the optimization approach. It was a single objective optimization in LION19 (we combine multiple factors in the same monodimensional objective function), solved via a solver. It is now multiobjective (considering passenger travel time, operation cost, and network similarity), and we now devise a modified NSGA-III algorithm.
- Because of the aforementioned differences, all results are new.

Finally, we confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission.

We believe the approach presented in this paper is novel and can have a strong impact both on the academic literature and, in the long term, on real PT operations.

Regards,
Assoc. Prof. **Andrea ARALDO**, **Zihao GUO** (PhD student), Prof. **Mounim EL-YACOUBI**
Institut Polytechnique de Paris

Dr. **Fayçal A. TOUZOUT**
National School of State Public Works

Contact: andrea.araldo@telecom-sudparis.eu